

Faculty of Science, Engineering and Built Environment

SEJ446 Engineering Project B

Deakin University Unit Guide

Trimester 2, 2025

CONTENTS

Welcome 2
Who is the unit team? 2
Unit chair: leads the teaching team and is responsible for overall delivery of this unit 2
Other members of the team and how to contact them 2
Administrative queries 2
About this unit 2
Unit development in response to student feedback 2
Learning Outcomes 3
Assessing your achievement of the unit learning outcomes 3
Hurdle requirements 3
Summative assessment task 1 4
Summative assessment task 2 4
Summative assessment task 3 5
Your learning experiences in this unit 5
Educator-facilitated (scheduled) learning activities - on-campus unit enrolment 5
Educator-facilitated (scheduled) learning activities - online unit enrolment 5
Typical study commitment 5
Note 6
Unit learning resources 6
Essential learning resources 6
Recommended learning resources 6
Key dates for this study period 6
Unit weekly activities 7

Welcome

This unit is a capstone project for students immediately prior to them leaving for their chosen career. Students will undertake project work proposed in SEJ440 or SEJ441, and then critically analyse and evaluate the results prior to communicating the project outcomes. It encompasses a number of technical engineering skills and an understanding of professional context in their discipline gained during the degree, and thus is a reflection of what they can anticipate when entering employment as a graduate engineer. Projects will have a research element to them in the justification of the approach and critical evaluation of results, although some may not necessarily be research-based, but rather industry specific. The learning style is predominantly student-led with direction from a supervisor through face-to-face or on-line meetings.

This Unit Guide provides you with the key information about this unit. Please read it carefully and refer to it frequently throughout the study period. Your unit site also provides information about your rights and responsibilities. We will assume you have read this before the unit commences, and we expect you to refer to it throughout the study period.

To be successful in this unit, you must:

- read all materials in preparation for your learning activities and follow up each with further study and research on the topic
- start your assessment tasks well ahead of the due date
- read or listen to all feedback carefully and use it in your future work
- attend and engage in all educator facilitated (scheduled) learning activities and other learning experiences as part of the unit design

Who is the unit team?

Unit chair: leads the teaching team and is responsible for overall delivery of this unit

Name: Ellen Moon
Campus: Geelong Waurn Ponds
Email: ellen.moon@deakin.edu.au
Phone: +61 3 522 78656
Other members of the team and how to contact them

You will continue to work with your supervisor from SEJ440 or SEJ441, as well as any Technical Officers in the relevant areas to your topic.
Administrative queries

- check-out the 'SEBE Student Hub' section on your unit site
- contact your Unit Chair or Campus Leader
- drop in or contact Student Central to speak with a Student Adviser

For additional support information, please see the Rights and Responsibilities section under 'Content' in your unit site.
About this unit

Unit development in response to student feedback

Every trimester, we ask students to tell us, through eVALUate, what helped and hindered their learning in each Unit. You are strongly encouraged to provide constructive feedback for this Unit when eVALUate opens (you will be emailed a link).
In previous versions of this unit, students have told us that these aspects of the unit have helped them to achieve the learning outcomes:

- Learning resources assisted in clearly understanding expectations and requirements.
- Zoom seminars assisted in providing clarity around assessment requirements and quality.

The following aspects of the unit have been introduced, enhanced or retained in response to feedback from students who

have undertaken this unit in previous trimesters:

- Provided more recent examples of previous students' assessment.
- The project management assessment has been embedded across the trimester rather than wholly left to the end of trimester to promote the regular use of project management tools and techniques, and better align with industry practice.

If you have any concerns about the Unit during the trimester, please contact the unit teaching team - preferably early in the trimester - so we can discuss your concerns, and make adjustments, if appropriate.

Learning Outcomes

Each Unit in your course is a building block towards Deakin's Graduate Learning Outcomes - not all Units develop and assess every Graduate Learning Outcome (GLO).

ULO	These are the Unit Learning Outcomes (ULOs) for this unit. At the completion of this unit, successful students can:	Alignment to Deakin Graduate Learning Outcomes (GLOs)
ULO1	Demonstrate advanced technical and theoretical knowledge in a specific area of their chosen discipline through the presentation, critical analysis and evaluation of project results.	GLO1: Discipline-specific knowledge and capabilities GLO2: Communication GLO4: Critical thinking GLO5: Problem solving
ULO2	Exhibit autonomy in conducting and managing a substantial project using formal project planning methods to a professional standard while accommodating stakeholder safety, risk and ethical requirements.	GLO6: Self-management GLO8: Global citizenship
ULO3	Demonstrate effective oral and visual communication skills in conveying project outcomes to a varied audience in an exhibition format.	GLO2: Communication
ULO4	Illustrate advanced technical written communication skills to the extent of producing a substantial formal report which conforms to specified formats and effectively communicates project outcomes.	GLO2: Communication GLO3: Digital literacy

Assessing your achievement of the unit learning outcomes

Hurdle requirements

To be eligible to obtain a pass in this unit students must demonstrate a satisfactory performance (equivalent to a pass) in each of the four unit learning outcomes (ULOs).

Brief summary of the hurdle requirement	Rationale
In this unit students must demonstrate a satisfactory performance (equivalent to a pass) in each of the four unit learning outcomes (ULOs).	The project completed in SEJ446 is the capstone of the Engineering course, and thus unit learning outcomes in this unit are directly aligned with the course learning outcomes that students are required to meet to obtain their degree.

Summative assessment (tasks that will be graded or marked)

NOTE: It is *your responsibility* to keep a backup copy of every assignment and the materials used to develop/complete it where possible (e.g. written/digital reports, essays, videos, images). In the unusual event that one of your submissions becomes corrupted, is incorrectly submitted or otherwise lost, you may be asked to submit the backup copy. Any work you submit may be checked by electronic or other means for the purposes of detecting breaches of academic integrity such as collusion, plagiarism and contract cheating. You must understand your responsibility to act with honesty and integrity in your studies as Deakin takes all breaches very seriously. Make sure you read Your rights and responsibilities as a student in this unit to find out more about academic integrity.

Deakin has a universal assessment submission time of 8 pm AEDT/AEST. A late penalty will apply to assessments submitted after 11.59 pm AEDT/AEST.

Summative assessment task 1

Title	**Project management and closeout**
Brief description of assessment task	This is an individual assessment task. Evidence should be provided that student has independently managed and executed the project to a professional standard and undertaken project work to a substantial depth and breadth, while showing appropriate consideration of stakeholder requirements and any safety/ethical considerations.
Detail of student output	Students are required to complete an online portfolio that includes regular status reports and evidence of project management activities (submitted regularly throughout the trimester), and culminates in a project closeout report that clearly articulates whether the project was achieved as planned, how the project will be handed over, how the student has met the assessment outcomes, and includes a reflection on learnings.
Grading and weighting (% total mark for unit)	30% marked
This task assesses your achievement of these Unit Learning Outcome(s)	This task directly assesses **ULO2.** By providing regular evidence of their project activities, stakeholder engagement and reflective skills, students demonstrate that they can conduct and manage a substantial project while accommodating stakeholder requirements **(ULO2)**.
This task assesses your achievement of these Graduate Learning Outcome(s)	Students will be assessed on how effectively and independently they have managed and conducted the project **(GLO6)** and how well they have addressed ethical and professional considerations **(GLO8).**
How and when you will receive feedback on your work	Students are expected to meet supervisors on a regular basis throughout the trimester and keep accurate minutes or logs of meetings, supervisor feedback and follow-up actions. Students will receive feedback on their portfolios during the trimester from the unit team.
When and how to submit your work	Status reports due in at regular intervals throughout the trimester as identified in the unit site. Final portfolio submission including closeout report due Friday Week 11 (26 September 2025) at 8pm (AEST) via the online portfolio tool in the unit site.

Summative assessment task 2

Title	**Project exhibition**
Brief description of assessment task	This is an individual assessment task. Students will present their work in an exhibition format and be assessed by a panel external to their supervisor based on clarity and creativity used in communicating the aims, scope and outcomes of their project.
Detail of student output	A short (5 minute) presentation followed by discussion and Q&A with multiple assessors. Students are expected to use visual aids, including a poster, as well as any other artefacts or visual aids developed or used by the student to assist in exhibiting and demonstrating their project.
Grading and weighting (% total mark for unit)	30% - marked
This task assesses your achievement of these Unit Learning Outcome(s)	This task directly assesses **ULO1 and ULO3**. By presenting their project work in an exhibition format, students demonstrate effective oral and visual communication skills **(ULO3).** During their presentations and subsequent discussions with assessors, students demonstrate advanced technical knowledge on their project topic **(ULO1)**.

This task assesses your achievement of these Graduate Learning Outcome(s)	Assessment panels will assess student's technical understanding **(GLO1)**, their oral and visual communication skills **(GLO2)** and their understanding of the importance, relevance and limitations of their project to industry and/or real-world applications **(GLO5)**
How and when you will receive feedback on your work	Students should seek feedback from their academic supervisor prior to the exhibition
When and how to submit your work	Week 12 - Students will be required to attend and present in-person at the Waurn Ponds campus. Exact times to be posted during trimester.

Summative assessment task 3

Title	**Final project thesis (report)**
Brief description of assessment task	This is an individual assessment task. Students need to produce a comprehensive technical report that captures the extent of their project, and includes a critical assessment of the relevant literature, a detailed methodology of the work undertaken, an accurate and detailed presentation of the results obtained, and a discussion of the relevance and implications of the results.
Detail of student output	The final report will conform to a traditional research thesis style (expected structure and headings will be conveyed to students through the unit site).
Grading and weighting (% total mark for unit)	40% marked
This task assesses your achievement of these Unit Learning Outcome(s)	This task directly assesses **ULO1 and ULO4.** By providing an in depth description of their project work and analysing and evaluating their project outcomes, students demonstrate advanced technical and theoretical knowledge in their discipline **(ULO1)**. By producing a substantial, well-written and well-structured thesis, students demonstrate their technical writing skills **(ULO4).**
This task assesses your achievement of these Graduate Learning Outcome(s)	The thesis will be assessed on student's technical knowledge **(GLO1)**, their written communication skills **(GLO2)**, their use of literature **(GLO3)** and the critical analysis of the literature and their own project results **(GLO4)**, and the general effectiveness of the project in addressing the stated project objectives **(GLO5).**
How and when you will receive feedback on your work	Students may submit a thesis draft by the end of Week 8 to allow sufficient time for academic supervisor to provide feedback on progress.
When and how to submit your work	End-of-unit assessment period - Thursday 9 October 2025 at 8pm (AEST) via the Dropbox in the unit site.

Your learning experiences in this unit

Educator-facilitated (scheduled) learning activities - on-campus unit enrolment

1 x 1 hour online seminar (weeks 1, 3, 5, 7 and 9).
In addition to Project work undertaken on campus, students are required to maintain a minimum of one hour fortnightly contact with their academic supervisor face-to-face and compulsory attendance on campus for the project exhibition on a single scheduled day at the end of the trimester. Refer to the unit site for further details.

Educator-facilitated (scheduled) learning activities - online unit enrolment

1 x 1 hour online seminar (weeks 1, 3, 5, 7 and 9).
Students are required to maintain a minimum of one hour fortnightly contact with their academic supervisor face-to-face or online and compulsory attendance on campus for the project exhibition on a single scheduled day at the end of the trimester. Refer to the unit site for further details.

Typical study commitment

Students will on average spend 300 hours over the teaching period undertaking the teaching, learning and assessment

activities for this unit.
This will include educator guided online learning activities within the unit site.
For campus and online students learning and assessment activities include online seminars, meetings with supervisor/s and technical staff, literature research, designated project activities and assessment tasks.

Depending on the project types, students may be required to conduct experiments and simulation. Access to all relevant hardware and software, that is required for the project and identified in SEJ440/SEJ441, will be provided.

Note

At Deakin, courses are delivered within a learning environment that provides all students with equitable and consistent access to facilities, infrastructure, resources, and support to assist student progress and achievement of learning outcomes.

We have introduced new terms to reflect learning activities to enhance your learning experience, aligning with our innovative **DeakinDesign learning principles and practices**. The new terms better reflect how teaching teams will guide you through your learning journey and the types of learning experiences you will have.

'Lectures' are the activities where teaching staff engage you through presentations with student participation.

In **'seminars'**, an educator will guide you in a smaller group of students through highly interactive discussions and activities.

Your units may also include '**practical experiences**' such as '**laboratory**', '**workshops**', '**clinical skills**' and more. These hands-on activities typically take place in specialised facilities with industry tools, equipment or technology to allow you to apply your knowledge practically.

Some other terms

If you see a '**meeting**' in your timetable, this is an optional drop-in session.

'Assessments' or **'team-based learning**' indicate an in-class evaluation of your skills or knowledge. A **'pre-assessment practice'** could be scheduled to prepare you for these assessments.

Find out more

Take a look at the **Learning activities webpage** for a full list of the terminology changes and reasons they were changed.
Unit learning resources

Your unit learning resources can be accessed from your unit site.
There is no prescribed text. Unit materials are provided via the unit site. This includes unit topic readings and references to further information.
Essential learning resources

There is no prescribed text for this unit.

Learning resources, including video recordings, that explains and details how students should approach their project and prepare for their assessment will be provided on the unit site.
Recommended learning resources

The Deakin Software Library provides students with access to software that you may need or find useful for your study at Deakin.

Useful learning sources will be discussed, as appropriate, with your supervisor and/or posted on the unit site during the trimester.
Key dates for this study period

Trimester 2 teaching period begins	Monday 7 July 2025
Census Date	Friday 15 August 2025
Intra-trimester break	Monday 11 August - Sunday 17 August 2025
Trimester 2 teaching period ends	Friday 26 September 2025
Study period (end-of-unit assessment/examination preparation period)	Monday 29 September - Friday 3 October 2025
End-of-unit assessment/examination period	Monday 6 October - Friday 17 October 2025
Inter-trimester break (the period between trimesters)	Monday 20 October - Friday 31 October 2025
Unit results released	Friday 31 October 2025 (10.30 am)

Unit weekly activities

Week	Commencing	Topic	Assessment due date
1	7 July 2025	Submit brief on your progress to your supervisor. Experimental (or other project methodology work) should commence. Establish a method of record keeping for project management report.	
2	14 July 2025	Experimental (or other) project work. Continue record keeping & reflection.	**Assessment 1: Project management and closeout** - Submit project status report
3	21 July 2025	Experimental (or other) project work. Continue record keeping & reflection.	
4	28 July 2025	Experimental (or other) project work. Continue record keeping & reflection.	**Assessment 1: Project management and closeout** - Submit project status report
5^	4 August 2025	Experimental (or other) project work. Continue record keeping & reflection.	
6	18 August 2025	Experimental (or other) project work. Continue record keeping & reflection.	**Assessment 1: Project management and closeout** - Submit project status report
7	25 August 2025	Finalise experimental work/result collection/build of design - commencement of analysis and preparation of draft thesis. Continue record keeping & reflection.	
8	1 September 2025	Analysis of results and preparation of draft thesis. Continue record keeping & reflection.	Submit draft thesis to supervisor **Assessment 1: Project management and closeout** - Submit project status report
9	8 September 2025	Prepare for project exhibition and finish any project work & critical analysis of outcomes. Continue record keeping & reflection.	
10	15 September 2025	Prepare for project exhibition and work on final thesis, taking on board supervisor feedback. Begin working on project close-out tasks within project management portfolio. Continue record keeping and reflection.	

Week	Commencing	Topic	Assessment due date
11#	22 September 2025	Project exhibition and thesis final preparation. Finalise report and consolidate records for project management assessment task.	**Assessment 1: Project management and closeout -** Final portfolio submission including closeout report due 26 September
12	29 September 2025	Project exhibition and thesis final preparation.	Study period **Assessment 2: Project exhibition** - Times for presentation will be posted on the unit site
13-14	6 October 2025		End-of-unit assessment and exam period **Assessment 3: Final project thesis (report)** due 9 October

^ Intra-trimester break: **Monday 11 August – Sunday 17 August 2025**
AFL Grand Final Eve public holiday: **Friday 26 September 2025** - University closed